

State of Michigan

Datacenter Substrate Alignment Guide

A Diagnostic Framework for Evaluating Large-Scale Compute Infrastructure

Prepared for: State-Level Infrastructure Review

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Executive Summary

Large-scale datacenters are increasingly central to economic development strategies. Their benefits are real, but so are their physical demands. Power, water, thermal output, land use, and community exposure form a shared substrate that cannot be scaled independently.

This guide introduces **Substrate Alignment** as a diagnostic lens for evaluating datacenter projects before irreversible commitments are made. It does not oppose development, nor does it propose regulation. Instead, it offers a structured way to surface second-order impacts that often emerge only after construction.

A Michigan-developed framework known as **Resonance-Time Theory / Vector Substrate Theory (RTT/vST)** is presented as a review tool capable of identifying cross-domain misalignment early, when options remain open.

Michigan has an opportunity to lead by applying substrate-aware review practices that protect long-term infrastructure coherence while supporting responsible growth.

1. What Is Substrate Alignment

Substrate refers to the physical and human systems that support all infrastructure:

- Electrical generation and transmission
- Water sourcing and thermal dissipation
- Land, noise, and heat exposure
- Workforce and surrounding communities
- Time horizons beyond initial deployment

Alignment occurs when system demands remain within the elastic limits of these substrates across time.

Misalignment does not usually appear at project approval. It appears later, as cumulative stress, escalating mitigation costs, and community friction.

Substrate alignment is not a policy position. It is a design condition.

2. Why Datacenters Stress Substrates Differently

Datacenters concentrate demand in ways most infrastructure does not:

- Continuous, non-interruptible power draw
- High thermal output requiring constant dissipation
- Water usage coupled to cooling strategies
- Local grid coupling that propagates upstream effects
- Long operational lifetimes with escalating capacity pressure

These characteristics make datacenters uniquely sensitive to substrate limits, especially when projects are accelerated.

3. Common Failure Modes Observed Nationally

Without naming specific projects, recurring patterns include:

Power Coupling

Local grids absorb initial load, then require regional reinforcement. Costs and impacts propagate beyond the host community.

Thermal and Water Feedback

Cooling strategies shift stress between air, water, and land, often trading one constraint for another.

Escalation Dynamics

Once built, capacity expansion becomes the default response to demand, compounding substrate stress.

Community Exposure

Noise, heat, traffic, and land use changes accumulate gradually, often outside original impact assessments.

These are not engineering errors. They are structural blind spots.

4. RTT/vST as a Diagnostic Review Tool

Resonance-Time Theory / Vector Substrate Theory (RTT/vST) was developed in Michigan to analyze systems that span multiple physical and human domains.

RTT/vST does not replace environmental review or engineering analysis. It complements them by:

- Mapping interactions across power, water, thermal, and social regimes
- Identifying resonance and saturation points before deployment
- Highlighting time-based escalation risks
- Making invisible coupling effects visible

It functions as a **coherence check**, not a compliance framework.

5. What a State-Level Substrate Review Could Look Like

A substrate alignment review can be lightweight and time-bounded:

- Conducted prior to final commitments
- Focused on power, water, thermal, and community coupling
- Informed by independent technical perspectives
- Designed to surface second-order effects, not halt projects

Such a review strengthens decision-making without delaying progress.

6. Michigan's Opportunity

Michigan has deep engineering talent, a diverse energy landscape, and a history of infrastructure leadership. By applying substrate-aware review practices, the state can:

- Reduce long-term infrastructure risk
- Protect communities and shared resources
- Set a national example for responsible scale

Substrate alignment is not about slowing development. It is about ensuring that growth remains coherent over time.

Closing

Infrastructure decisions shape the physical and social field for decades. Making that field visible before it is altered is an act of stewardship.

This guide is offered as a reference, not a directive.